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〔Review Article〕

New approaches in the design of capillary electrophoresis experiments

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SUMMARY

This paper gives a survey of recent methodological studies of capillary electrophoresis at the Department of Biochemistry, Uppsala University. Several methods for on-and off-tube concentration of solutes have been developed. These methods also permit desalting of the sample, which is necessary for the creation of narrow starting zones in zone electrophoresis and to reduce the risk of precipitation and narrowing of the separation window in isoelectric focusing (IEF). A unique method designed particularly for desalting of samples for IEF is also discussed. To obtain high resolution the adsorption of solutes to the capillary wall must be eliminated. A pH stable, hydrophilic polymer coating has, therefore, been developed. Although polymer solutions give lower resolution than do gels they are often used as molecular-sieving media because they are replaceable and thus permit repetitive automated analyses in the same capillary. We have shown that low-melting methoxylated agarose gels are also replaceable and give the same high resolution as do polyacrylamide and dextran gels. Following an electrophoresis the methoxylated agarose gels can be displaced in the stationary capillary past the detection window. This scanning technique has several advantages. A capillary with strong electroendosmosis permits, in principle, separation of both acidic and basic proteins in one run. However, depending on whether the capillary wall is negatively or positively charged either the basic or the acidic proteins most often migrate with strong tailing caused by the electrostatic interaction with the capillary wall. The distortion of the zones is much less if the electroendosmotic flow is replaced by a hydrodynamic flow and the electrophoresis is performed in a coated electroendosmosis-free capillary to suppress adsorption of both basic and acidic proteins. By a unique technique we can prepare chromatographic beds with diameters as small as 5-25 μm. These beds, which are more homogeneous than conventional beds built up of preformed beads, can be used with advantage for electrochromatography and for different modes of capillary chromatography of fractions obtained from micropreparative HPCE experiments. There is a general trend and desire to decrease the analysis time without sacrificing resolution.

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Therefore, we have developed low-conductivity buffers with satisfactory buffering capacity which permit field strengths as high as 2,000 V/cm with attendant very short run times. Many enantioselective agents gave strong UV-absorption. Therefore, the experimental conditions must be chosen so that they do not pass the detection window. Otherwise, the noisy background is very disturbing. When a zone migrates from a straightinto a curved section of a capillary the molecules at the outer lane will lag behind those at the inner lane. The attendant loss in resolution, which all commercial apparatus exhibits, can be decreased considerably by coiling the capillary into the figure-of-eight or a serpentine path. A straight capillary is, however, preferable. Methods for the determination of pH and electrical conductivity in small volumes are discussed. A technique for micropreparative HPCE and HPLC is presented. An HPCE apparatus designed according to new principles is described (straight capillary; one electrode vessel is closed to avoid hydrodynamic flow in the capillary; electroendosmotic pump for washing of the capillary; thermostated slit for the UV beam to obtain a straight base line; only a small part of the capillary is not cooled actively; thermal application of the sample; the sample rests as a small droplet on a carousel; the droplet is covered by a cap which can be lifted by a magnet; effective insulation for field strengths up to at least 2,000 V/cm).

Key words: capillary electrophoresis, on-tube concentration, off-tube concentration, desalting, coating, low-conductivity buffer, electrochromatography, low-melting agarose, dextran gel, enantiomer separations, coiled capillaries, gel scanning, pH, electrical conductivity, capillary electrophoresis apparatus.

INTRODUCTION

High-performance capillary electrophoresis (HPCE) is a widely used separation method because it permits short analysis times, affords high resolution and requires only small amounts of sample. However, to utilize the full potential of HPCE methodological improvements have to be introduced. This paper deals with novel methods recently developed at the Department of Biochemistry, Uppsala University, and presented at the 45 th general meeting of the Japanese Electrophoresis Society. These methods are described below under separate titles to facilitate for each reader to con- centrate on those aspects which are of particular interest.

ON-TUBE CONCENTRATION

UV-detection in HPCE requires relatively high sample concentrations due to the small diameter (light path) of the capillary. Therefore, we have developed a series of methods for 20-1,000-fold concentration of the solutes.[1,2]

The electrophoretic migration velocity v of a solute can be described by the equation:

$$v=\mu\cdot\frac{I}{q\cdot\varkappa} \qquad (1)$$

where μ=the mobility, I=the current, q=the cross-sectional area of the electrophoresis tube and $\varkappa$=the conductivity.

Assume that we use experimental conditions such that the front of the applied starting zone migrates more slowly than the rear. The width of the zone then decreases continuously, i.e., a concentration of the sample is achieved. Eq. (1) shows that this can be accomplished by having the zone migrate toward: 1) a gel with pores too small for the solutes to penetrate ($\mu\rightarrow 0$), 2) a pH gradient ($\mu\rightarrow 0$ for ampholytes), 3) a solution with large cross-section q, or 4) a solution with very high conductivity, $\varkappa$. These four methods are described in Refs. 1) and 2) along with a fifth method based on a combination of displacement electrophoresis and hydrodynamic counterflow. An example of the latter approach is shown in Fig. 1 a. The sample consisted of β-lactoglobulin, α-lactoalbumin and hemoglobin A, each protein dissolved at a concentration of 5 μg/ml in 0.015 M HCl-Tris, pH 8.5 (leading buffer). The anode vessel was filled with the same buffer and the

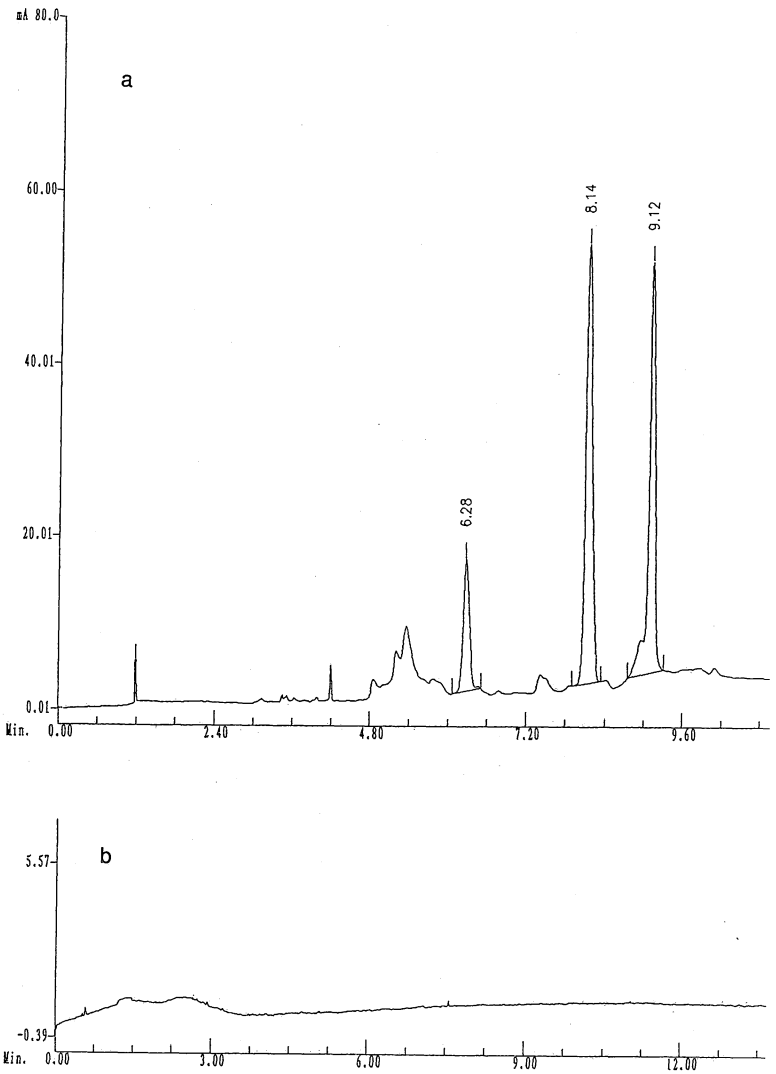


Fig. 1. HPCE of proteins following on-tube concentra- tion.

The whole capillary was filled with the sample solution, which was concentrated to a 0.2 mm wide starting zone by displacement electrophoresis in combination with a hydrodynamic counterflow. The analysis was then performed as a free zone electrophoresis (a). The concentration was 1,000-fold (the original concentration of each protein was only 5 μg/ml). A series of automated runs can be performed in commercial HPCE apparatus (cf Fig. 5). The result of a blank experiment without concentration of the sample is shown in (b).

cathode vessel with 0.01 M glycine-NaOH, pH 8.8 (terminator). The molarities refer to the first-mentioned buffer constituent. The capillary with the dimensions 0.1 mm (I.D.) $\times$240 mm was coated with linear polyacrylamide.[3] The entire capillary was filled with the sample solution. The analysis

was performed at 8,000 V and 15°C. The buffer level in the anode vessel was adjusted to a height which was 18 mm higher than that in the cathode vessel in order to create a hydrodynamic flow in the capillary with a velocity equal to the electrophoretic velocity of the terminating (glycine) ions, i.e., the proteins became sandwiched between the backward migrating leading (chloride) and the stationary terminating (glycine) ions by displacement electrophoresis. At the steady state the total width of the protein zones was about 0.2 mm, as estimated in a separate experiment with colored proteins. The displacement electrophoresis step was transferred to a zone electrophoresis step by exchanging the terminator in the cathode vessel for the leading buffer (Fig. 1 a). Since the length of the capillary was 240 mm and the width of the starting zone was 0.2 mm the sample was concentrated $240/0.2 \approx 1{,}000$ fold. A free zone electrophoresis experiment with the same sample but without concentration is shown in Fig. 1 b. A com- parison with Fig. 1 a shows the efficiency of the concentration method.

The experiment was performed in BioFocus 3000 from Bio-Rad Laboratories, Hercules, USA. This apparatus permits automation of the entire con- centration procedure.

OFF-TUBE CONCENTRATION

This method is based on the use of a "gel syringe", prepared in the following way. A sealed, vertical Pasteur pipette, into the tip of which a fused silica tube (O.D. about 0.4 mm, length about 150 mm) has been inserted, is filled with a solution of acrylamide, bisacrylamide and a catalyst system. Following polymerization the sample is sucked into the "gel syringe" by displacing the fused silica tube (the piston) backwards. The gel should have such small pores that only small ions, but not peptides and proteins, can penetrate it. Accordingly, small ions will leave the sample by diffusion into the gel, i.e., a desalting takes place at the same time as the larger molecules are concentrated by the Donnan effect (water is sucked out of the sample solution). The desalting takes about 15 min and the concentration about 30 min. For details, see Ref. 4). The method permits a 20-fold concentration. However, if the gel is prepared in the electrophoresis buffer diluted 1 : 10 with water, the concentration of the sample following the electrophoretic zone sharpening will be almost $10 \times 20 = 200$ fold.

We are developing a simple off-tube concentration method, based on a thin-walled hollow fiber made, for instance, of cellulose acetate. The fiber (I.D. $\approx$ 0.2 mm) is filled with the sample, the molecular weight of which should be so large that the sample molecules cannot penetrate the pores in the fiber wall. One end of the fiber is closed. The small water molecules evaporate quickly through the pores in the fiber wall. At the same time the sample solution moves toward the closed end where it is collected in concentrated form. Salts can then be removed rapidly by dipping the fiber end in water.

DESALTING

With the above on-and off-tube concentration methods the sample can be desalted efficiently. This is essential because a sample zone with a conductivity higher than that of the surrounding will broaden in the initial phase of the electrophoresis. For high resolution in isoelectric focusing it is also necessary to desalt the sample, for instance, by one of the methods described. However, we have designed a simple desalting technique especially for isoelectric focusing, that involves addition of carrier ampholytes to the anolyte and catholyte.[5]

SUPPRESSION OF ELECTROENDOSMOSIS AND ADSORPTION

Electroendosmosis has its origin in charges on the capillary wall. Therefore, any capillary that causes electroendosmosis also exhibits electrostatic interactions with solutes of a charge opposite to that of the capillary wall. Consequently, any method which suppresses electroendosmosis also decreases electrostatic adsorption of solutes. A simple method is to add a non-ionic, hydrophilic polymer, for instance, methyl cellulose, to the buffer[6,7]— a technique which has experienced a revival during recent years. A better technique is to coat the wall with such a polymer. The viscosity in the double layer is then so high that the electroendosmosis is virtually zero.[7] The methods

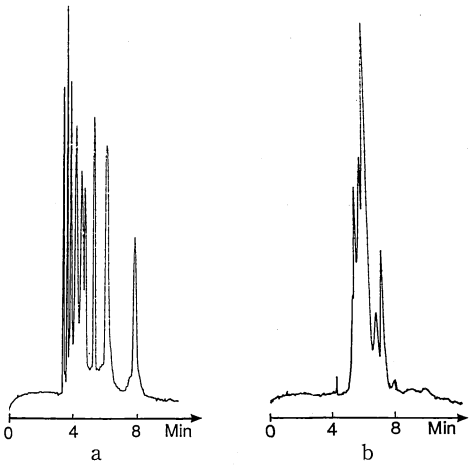


Fig. 2. HPCE of model proteins in a methyl cellulose coated capillary.

Sample : in a) : cytochrome C, lysozyme, myoglobin, conalbumin, carbonic anhydrase, RNase, α-chymotrypsinogen A, α-lactalbumin, and soya bean trypsin inhibitor. In b) : ovalbumin. The experiments in a) and b) were performed at pH 3.5 and 7.5, respectively. In a) the proteins had a positive charge and in b) a negative charge. The adsorption of the proteins onto the capillary wall is negligible in both experiments since the peaks are symmetrical.

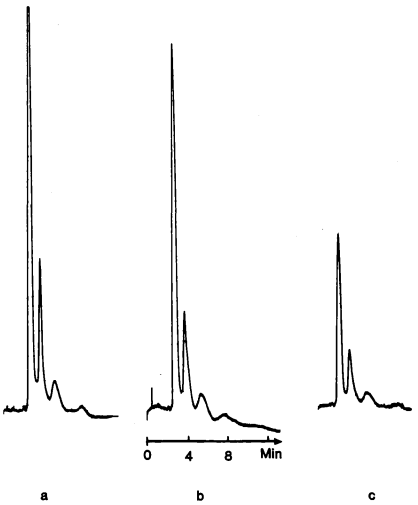


Fig. 3. HPCE of serum albumin in a polyacrylamide gel.

The detection was done at a) 230, b) 220 and c) 210 nm. The peaks are much lower at 210 nm than at 230 nm, although proteins have higher absorption at the lower wavelength. This is caused by the very high UV absorption of the polyacrylamide gel, which hinders the detection of low protein concentrations (compare Fig. 4, where the advantage of using dextran gels is demonstrated).

described in Refs. 3) and 8) are based on this approach and refer to coatings of acrylamide and methyl cellulose, respectively, the latter being more stable at high pH.[9] A test for the inertness of a coating is to analyze a sample of positively charged solutes (since silica contains negative silanol groups). Such a run is presented in Fig. 2 a, which demonstrates the absence of adsorption, since all of the protein peaks are symmetrical. The high resolution in the coated capillaries is demonstrated in Fig. 2 b, which shows that proteins with small differences in mobility can be resolved.

MOLECULAR-SIEVE ELECTROPHORESIS IN METHOXYLATED AGAROSE

Polyacrylamide gel electrophoresis has been widely used since its introduction around 1960, mainly due to the high resolution. However, it has obvious drawbacks when used in the capillary format. Particularly disturbing is the high UV absorption of polyacrylamide gels, which precludes the possibility to detect low protein concentrations at short wavelengths. The toxicity of the monomers and the risk of formation of water vapor bubbles in the gel during the electrophoresis are additional disadvantages.

The first disadvantage is illustrated in Fig. 3. The gel had the composition $T=6\%$ $T=3\%$ (these parameters are defined in Ref. 10)). Sodium borate (0.05 M), pH 9.0, was used as buffer. The sample consisted of human serum albumin dissolved in the buffer, diluted 1 : 20, to a concentration of 5 mg/ml. The analysis was performed at 1,000 V and the detection at 230, 220 and 210 nm. The peaks in Fig. 3 correspond to monomers, dimers, trimers, and tetramers of albumin. The absorption coefficient of proteins increases with a decrease in wavelength from 230 to 210 nm. However, the heights of the peaks are lower at 210 than at 230 nm because of the strong UV absorption of the gel.

To suppress this disturbing UV absorption we have replaced the polyacrylamide gel by a dextran gel prepared as follows. At high pH dextran was derivatized with allyl glycidyl ether

to produce allyl dextran.[8] Acrylamide (5 mg), TEMED (0.25 μl), and potassium persulfate (1 mg) were added to 1 ml of a 9% (w/v) solution of allyl dextran in 0.02 M sodium phosphate, pH 7.5. This mixture was sucked into a 0.075 mm (I.D.) × 105 mm fused silica tubing coated with linear polyacrylamide.[3] Following polymerization overnight a sample of human serum albumin dissolved in the buffer (diluted 1 : 4) at a concentration of 1 mg/ml was applied electrophoretically at 1,500 V for 5 sec. The electropherogram developed at 1,500 V and at a wavelength of 210 nm is presented in Fig. 4 a. A similar experiment was performed in a polyacrylamide gel of the composition $T=6\%$, $C=3\%$ (Fig. 4 b). A comparison of these figures shows that dextran gels permit detection of considerably lower protein concentrations than do polyacrylamide gels (in spite of the fact that the sensitivity of the detector in the polyacrylamide gel experiment was 2.5-fold higher than that in the dextran gel experiment and 30% more material was applied). Both the polyacrylamide and the dextran gels have a limited lifetime, particularly in the analysis of proteins. Therefore, they have been replaced by molecularsieving polymer solutions.[11-15] The advantage of such a solution is that it can easily be replaced after a run by a fresh polymer solution for the analysis of another sample, and so on. However, this practical advantage is gained at the cost of some resolution.[16,17] Therefore, we have searched for a replaceable *gel* with the same resolution as polyacrylamide gels.[18] The latter gels have the structure of crosslinked random coils. One can therefore expect that any high-resolving gel must have this conformation. It should be possible to prepare an agarose gel with such a structure by preventing the formation of hydrogen bonds between the galactose chains, which can be accomplished by derivatizing the galactose units in the agarose chains with methoxy groups, for instance. Such agarose gels, which were synthesized for us by Dr. Rafael Armisen, Hispanagar, Burgos, Spain, have a low gelling temperature (12-27°C). An automated series of experiments in the BioFocus 3000 with such an agarose gel is shown in Fig. 5. A 1% solution of methoxylated agarose (gelling temperature 25.6°C) was pressed from a vial in the carousel (thermostated at 40°C) into the fused silica tubing (0.075 (I.D.) × 240 mm) coated as described in Ref. 3) to suppress electroendosmosis (which otherwise displaces the gel out of the tubing). The temperature of the coolant was lowered from 35°C to 15°C to accomplish gelation of the agarose solution. The analysis of the sample was then performed in a conventional way at constant current of 5.80 μA (3,000 V). The gel was pressed out following the analysis and replaced by a fresh agarose solution, and so on. We have used the BioFocus 3000 for these automated runs since this apparatus gives the pressure required (2-7 bar) to displace the gel. The very high resolution in replaceable agarose gels demonstrated in Fig. 5.

An example of a separation of polydeoxyadenylic acids in these gels is shown in Fig. 6.

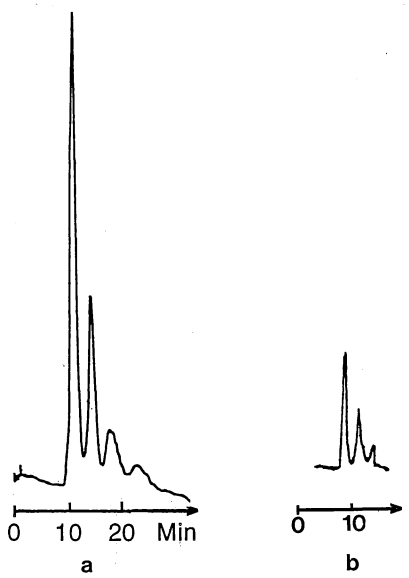


Fig. 4. A comparison of the electropherograms from HPCE experiments in a dextran (a) and a polyacrylamide (b) gel.

Sample: human serum albumin. Notice that dextran gels afford much higher peaks than do polyacrylamide gels, indicating that dextran gels are considerably more permeable to UV light and therefore permit analysis of smaller amounts of proteins. In fact, the peaks in a) would have been 3-fold higher if the experimental conditions had been identical (see text).

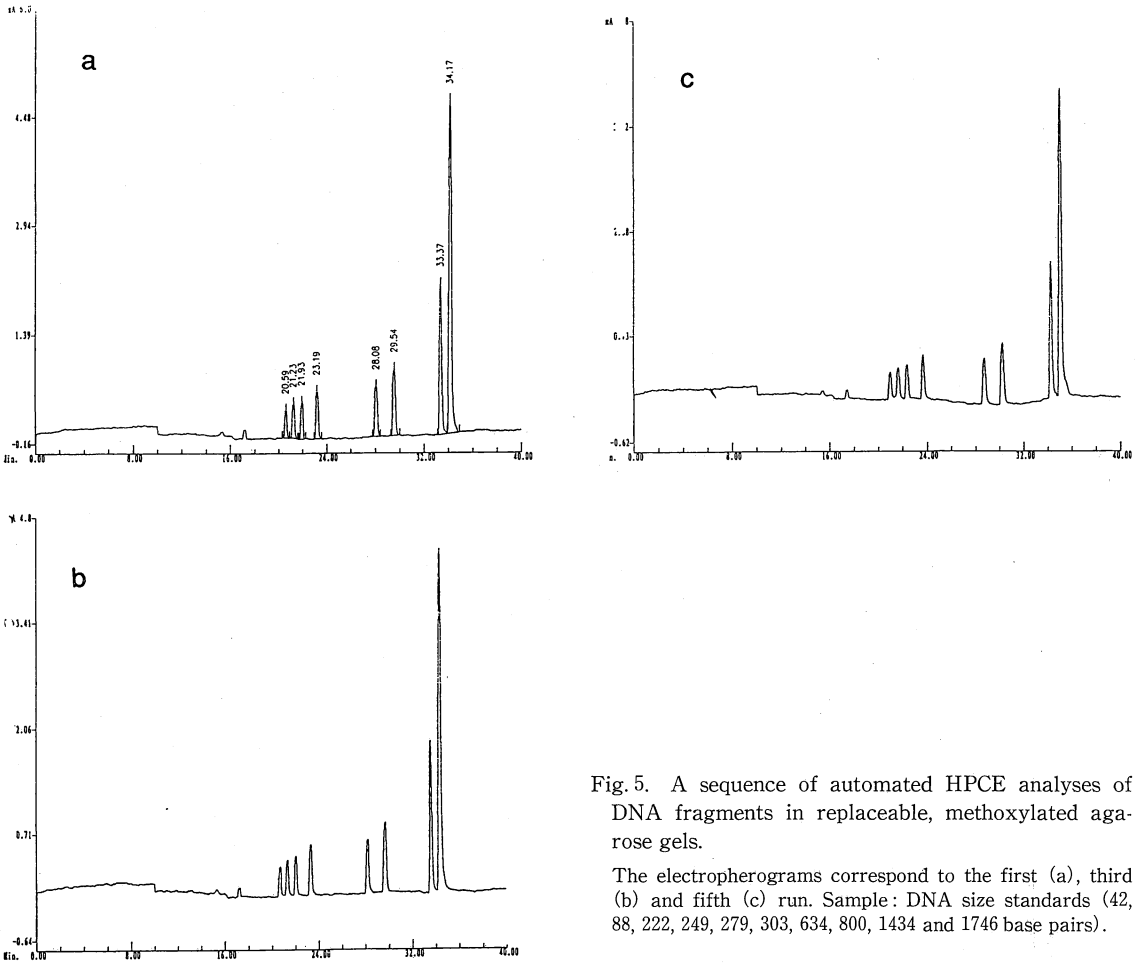


Fig. 5. A sequence of automated HPCE analyses of DNA fragments in replaceable, methoxylated agarose gels.

The electropherograms correspond to the first (a), third (b) and fifth (c) run. Sample: DNA size standards (42, 88, 222, 249, 279, 303, 634, 800, 1434 and 1746 base pairs).

SCANNING OF A GEL FOLLOWING AN ELECTROPHORETIC ANALYSIS

Methoxylated agarose with a gelling temperature of 25.6°C was dissolved at a concentration of 1% (w/v) in 0.04 M Tris-acetate, pH 7.5, containing 1% (w/v) SDS, using a water bath at 90°C. The hot agarose solution was sucked into a polyacrylamide-coated piece of fused silica tubing (0.05 (I.D.) × 318 (277) mm). The gelation took place overnight at room temperature. The sample (human serum albumin, 1 mg/ml) was applied electrophoretically (2,500 V, 10 s). The electrophoresis was conducted at 2,500 V for 20 min (this time was not sufficient for the solute zones to reach the detection window). With the voltage switched off the gel was then forced out of the capillary using a syringe pump. The scanning pattern is presented in Fig. 7. The detection was done at 210 nm. The agarose gels have the advantage over polyacrylamide gels that they are UV-transparent and can easily be pressed out of the capillary. This scanning technique permits detection of very slowly migrating components. It also gives information about possible precipitates formed during the run, since they appear as sharp spikes in the electropherogram. In addition, by moving the gel back and forth during a run one can follow the separation during the whole analysis. It should be emphasized that the displaced gel moves in the capillary as a rigid body with no distortion of the zones, which is in sharp contrast

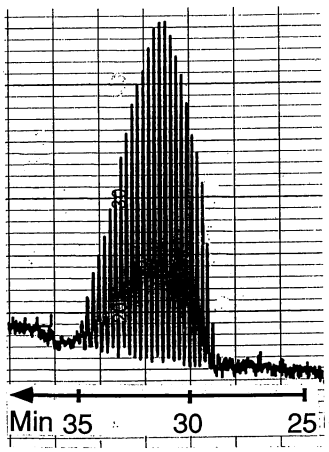


Fig. 6. HPCE analysis of a polydeoxyadenylic acid mixture in methoxylated agarose gel.

Sample: poly $(dA)_{40-60}$. Buffer: 0.04 M Tris-acetic acid, pH 8.4, containing 2 mM EDTA, 32% formamide and 5.6 M urea. The gelling point of the agarose was 20.8°C. The dimensions of the polyacrylamide-coated capillary: 0.1 (I.D.) × 300 (255) mm. Voltage: 6,000 V (16 μA).

to the parabolic deformation of a zone upon hydrodynamic displacement following a carrier-free zone electrophoresis.

DETECTION OF ACIDIC AND BASIC ANALYTES IN THE SAME RUN IN A COATED (ELECTROENDOSMOSIS-FREE) CAPILLARY USING A HYDRODYNAMIC FLOW

Electrophoresis in a coated electroendosmosis-free capillary does not permit analysis of both acidic and basic solutes in the same run, as is illustrated in Fig. 8 a, I, where the basic solute migrates out of the capillary.

Since in an uncoated capillary the electroendosmotic mobility is usually higher than the electrophoretic mobility of the acidic solute the latter will not migrate backward out of the capillary, but, unfortunately, the basic (positively charged) solute will interact electrostatically with the negative groups on the capillary wall, which causes zone broadening (tailing) as illustrated in Fig. 8 a, II.

However, by using a hydrodynamic flow in a coated electroendosmosis-free capillary the adsorption of the basic solute can be eliminated and

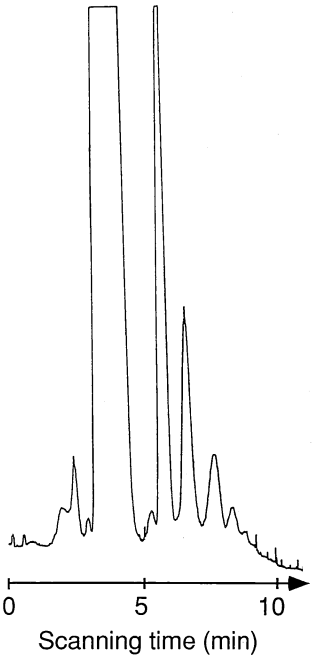


Fig. 7. Scanning of a gel of methoxylated agarose.

Sample: serum albumin. During the electrophoresis step, which took 20 min, the proteins did not reach the detection window. In the subsequent scanning step the rigid gel with the protein zones was moved past the detection window with the aid of a syringe pump (the voltage was switched off).

the acidic solute prevented from leaving the capillary (Fig. 8 a, III). The hydrodynamic flow causes a parabolic deformation of the zones. This deformation strengthens (weakens) the parabolic, thermal deformation of zones containing basic (acidic) solutes. At some particular hydrodynamic flow the zone with the acidic solute is not distorted (Fig. 8 a, III). Fig. 8 b shows an analysis in which this approach was utilized.

ELECTROCHROMATOGRAPHY ON CONTINUOUS BEDS

There is an ever increasing need for analysis techniques on a micro scale. Therefore we are studying methodologically not only capillary electrophoresis, but also microchromatography on columns with inner diameters in the range 5-300 μm. These columns are not packed in the conventional way with preformed beads, but consist of a continuous polymer rod with channels through

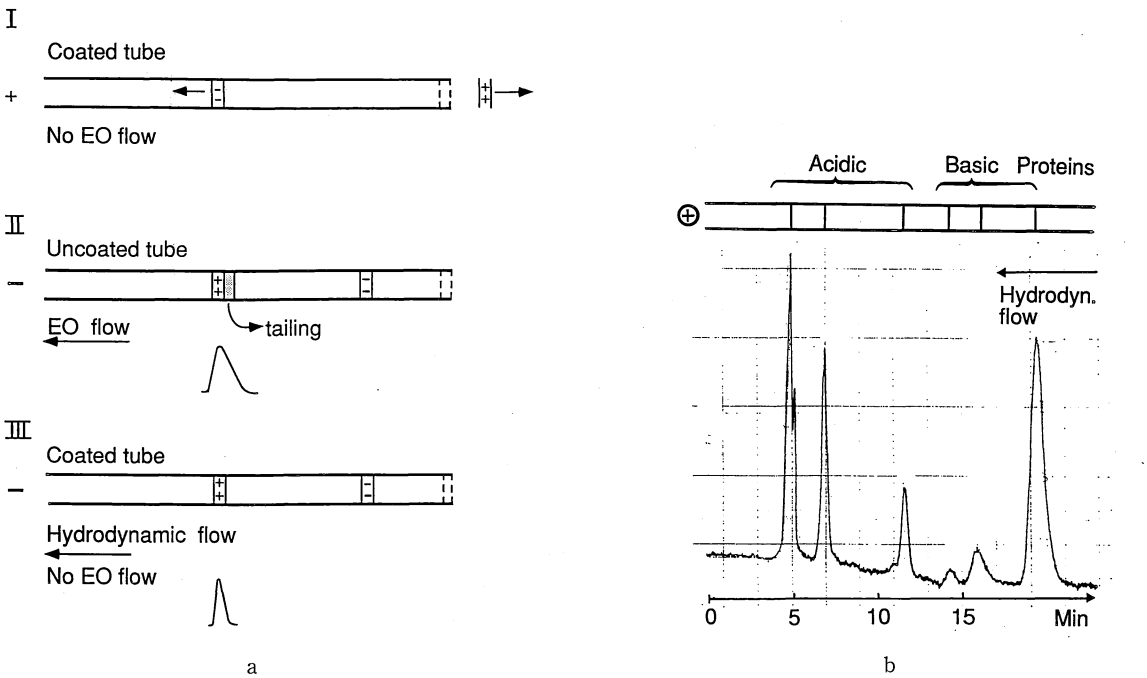


Fig. 8. Simultaneous detection of both acidic and basic proteins.

a) The principle. Basic proteins do not enter an electroendosmosis-free capillary (I). Both basic and acidic proteins enter an uncoated capillary with strong electroendosmosis, but the basic proteins become adsorbed (II). Both basic and acidic proteins enter a coated electroendosmosis-free capillary, although the zones become somewhat distorted by the hydrodynamic flow (III). However, the acidic solute zone is undistorted when the patabolic deformation caused by the counterflow is equal to the parabolic deformation caused by the Joule heat.[7] b) An actual experiment with three acidic and three basic proteins. Sample: β-lactoglobulin, human serum albumin, transferrin (acidic proteins) and ribonuclease, cytochrome C and a contaminant (basic proteins). Buffer: 0.1 M sodium phosphate, pH 7.3. Dimensions of the capillary: 0.05(I.D.) × 125 mm, Voltage: 3,600 V. Detection wavelength: 220 nm.

which the eluent can pass. These so-called continuous beds are prepared by a simple and costeffective procedure: a solution of appropriate monomers in the presence of salt (or a polymer) is sucked into a chromatographic tube (fused silica tubing) and is allowed to polymerize. The bed becomes attached covalently to the wall of the chromatographic tube during the polymerization. An example of an electrochromatographic separation of neutral solutes in a continuous bed is shown in Fig. 9. The bed was prepared from hydroxyethyl methacrylate, methacrylamide, vinylsulphonic acid, allylglycidyl ether, and piperazine diacrylamide and then derivatized with C_{18}- groups.

ELECTROPHORESIS AT HIGH FIELD STRENGTHS IN LOW-CONDUCTIVITY BUFFERS

The velocity, v, of a solute is determined by Eqn (1). Upon dilution of a buffer, $\varkappa$ decreases and μ increases which is favorable for high velocities, i.e., for short analysis times. However, the buffering capacity decreases when the buffer

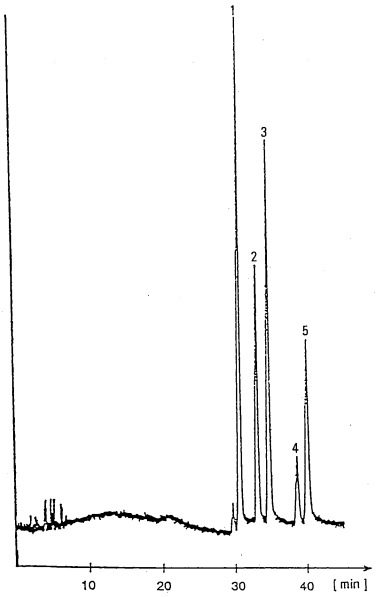


Fig. 9. Electrochromatography on a continuous bed.

The dimensions of the capillary: 0.025 (I.D.) × 660 mm. Sample: 1, naphthalene; 2,2-methylnaphthalene; 3, fluorene; 4, phenantrene; 5, anthracene. Mobile phase: 4 mM potassium phosphate, pH 7.8, containing 60% (v/v) acetonitrile.

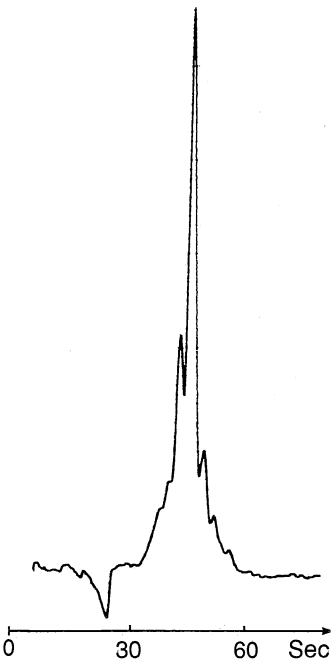


Fig. 10. HPCE at high field strength in a low-conductivity buffer.

Sample: transferrin (2 mg/ml). Dimensions of the coated capillary: 0.05 (I.D.) × 160 mm. Buffer: 0.01 M glycidylhistidine, pH 7.5. Detection at 210 nm. Voltage: 30,000 V (2 μA).

concentration is lowered, which sets a practical limit to how far a buffer can be diluted. Ideal buffers are those in which both (all) buffer constituents contribute to the buffering capacity and have a low mobility and valency to give a low conductivity ($\varkappa = F/1{,}000 \cdot \Sigma c\mu$, where F = the Faraday constant, c = the solute concentration in equivalents per litre and μ = the mobility). Ampholytes at their isoelectric points, where they have very low conductivity, should also be appropriate buffer substances. We have designed several buffers along these lines.[19] An example is glycidyl histidine. When this dipeptide in its free form is dissolved in carbon dioxide free water the pH will become close to the pI (=7.5) of the peptide. At this pH both of the basic groups have satisfactory buffering capacity (pK=6.8 and 8.2). A free zone electrophoresis of transferrin in a low-conductivity buffer (glycidyl histidine) is shown in Fig. 10. At a field strength of 1,900 V/cm (2.1 μA) the run was completed in one min.

Since these low-conductivity buffers pass very little current the Joule heat is low ($1{,}900 \times 2.1 \times 10^{-6} = 0.04$ Watt in the above experiment), which means very little thermal zone broadening. The broadening caused by diffusion is also small, since the running time is so short (the diffusional broadening of a peak at the base line in Fig. 10 is about $4\sqrt{2\,Dt} \approx 0.1$ mm.

Since both the thermal and diffusional broadenings are small in the low-conductivity buffers the resolution can be very high if adsorption of the solutes to the capillary wall can be eliminated. Therefore, it is important that the capillaries are coated efficiently, for instance, as described in Refs. 3), 8) and 20).

The minimum concentration of a low conductivity buffer should be determined by running the sample in buffers of increasing dilution and observing at which dilution the resolution begins to decrease.

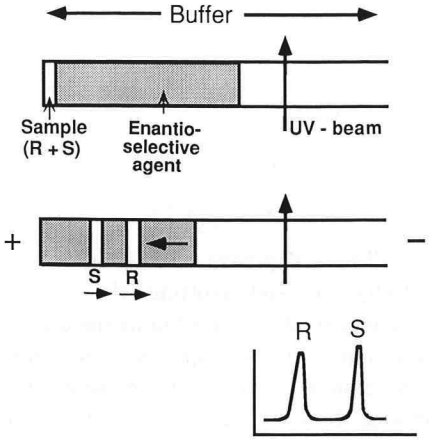


Fig. 11. Separation of enantiomers by HPCE in the presence of a UV-absorbing enantioselective agent.

The experimental conditions are chosen such that these agents do not pass the detection window at the same time as the UV-absorbing enantiomers.

SEPARATION OF ENANTIOMERS

For the separation of R-and S-forms an enantioselective agent is dissolved in the buffer. This procedure has been used also when the agent is a protein. However, upon UV-detection the base line is very irregular due to the strong UV absorption of proteins. To avoid these disturbances one should choose such experimental conditions that the protein (or any other UV-absorbing enantioselective agent) either migrates in a direction opposite to that of the enantiomers or migrates slower or faster, so that the protein zone will not be at the detection window at the same time as the enantiomers and the base line will be straight. The first alternative is illustrated in Fig. 11.

THE INFLUENCE OF THE SHAPE OF THE CAPILLARY ON THE RESOLUTION

It is known that a straight capillary gives a higher resolution than does a coiled capillary.[21,22] In the absence of diffusion the zone broadening is equal to $n2\pi\phi$, where n=the number of loops and ϕ=the diameter of the capillary.[21,22] However, by coiling the capillary into a figure-of-eight (Fig. 12 a) or into a serpentine path (Fig. 12 b) the broadening is strongly suppressed (Fig. 13).

Following polymerization of acrylamide mono-

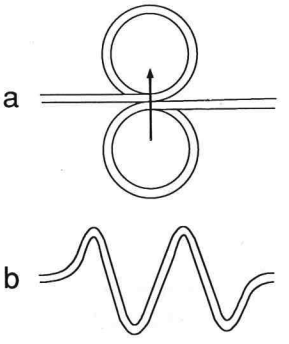


Fig. 12. Approaches to suppress the zone broadening caused by the curvature of the capillary.

a) The capillary bent into a figure-of-eight. b) The capillary bent into a serpentine path.

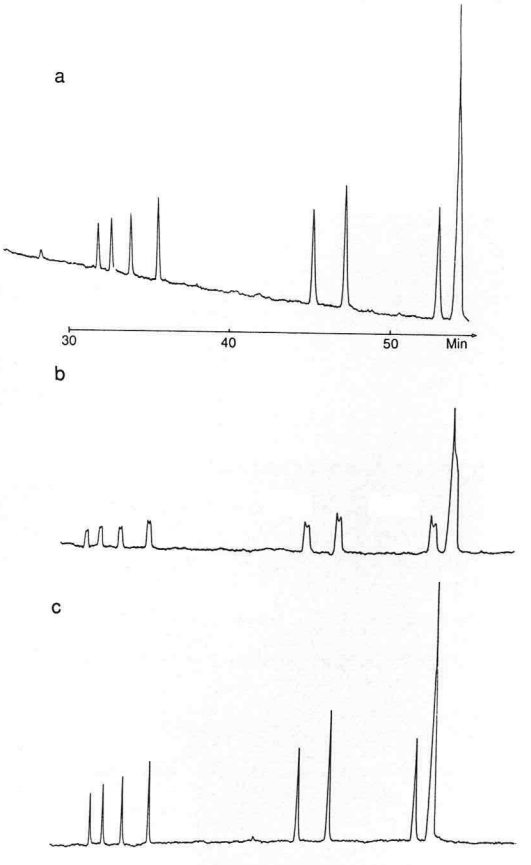


Fig. 13. A comparison of the resolution in polyacrylamide gel-filled capillaries bent into different shapes (compare Fig. 12).

Sample: DNA size standards (see Fig. 5). The run was done in a) a capillary bent into figure-of-eight, b) in a capillary bent into a loop and c) in a straight capillary.

mers in a capillary bent into figure-of-eight (Fig. 13 a), the capillary was stretched out into a loop (Fig. 13 b) and then into a straight configuration (Fig. 13 c). The decrease in resolution in Fig. 13 b is partly influenced by the heterogeneities in the gel introduced upon bending (not shown herein).

Observe that the experiment in Fig. 13 was performed in a gel of polyacrylamide. In a carrier-free zone electrophoresis the lateral diffusion is more pronounced, which decreases considerably the difference in the observed, total zone broadening between a straight and a coiled capillary.

Notice that the configuration in Fig. 13 a permits recording of three consecutive electropherograms during a single run.[21] An additional advantage is that mobilities and relative peak areas can be determined more accurately than in a straight capillary.

DETERMINATION OF pH AND ELECTRICAL CONDUCTIVITY IN SMALL VOLUMES AND OF THE EFFECTIVE DIAMETER OF A CAPILLARY

Since $R=L^1/\varkappa q$ Ohm's law can be written

$$V=\frac{4\,L^1}{\varkappa\pi\phi^2}\cdot I \qquad (2)$$

Upon filling a capillary with a buffer of known conductivity, $\varkappa$, and plotting the voltage, V, against the current, I, one obtains the diameter, ϕ, from the slope of the straight line, since the length L^1 of the capillary can be determined easily. After determining the effective diameter of the capillary in this way the conductivity of any sample, buffer, etc., can be calculated in a similar manner.

Most pH indicators have two absorption maxima. The ratio between the absorption values at these maxima changes with the pH of the solution. From a plot of this ratio against pH it is easy to obtain the pH of a sample. A modern

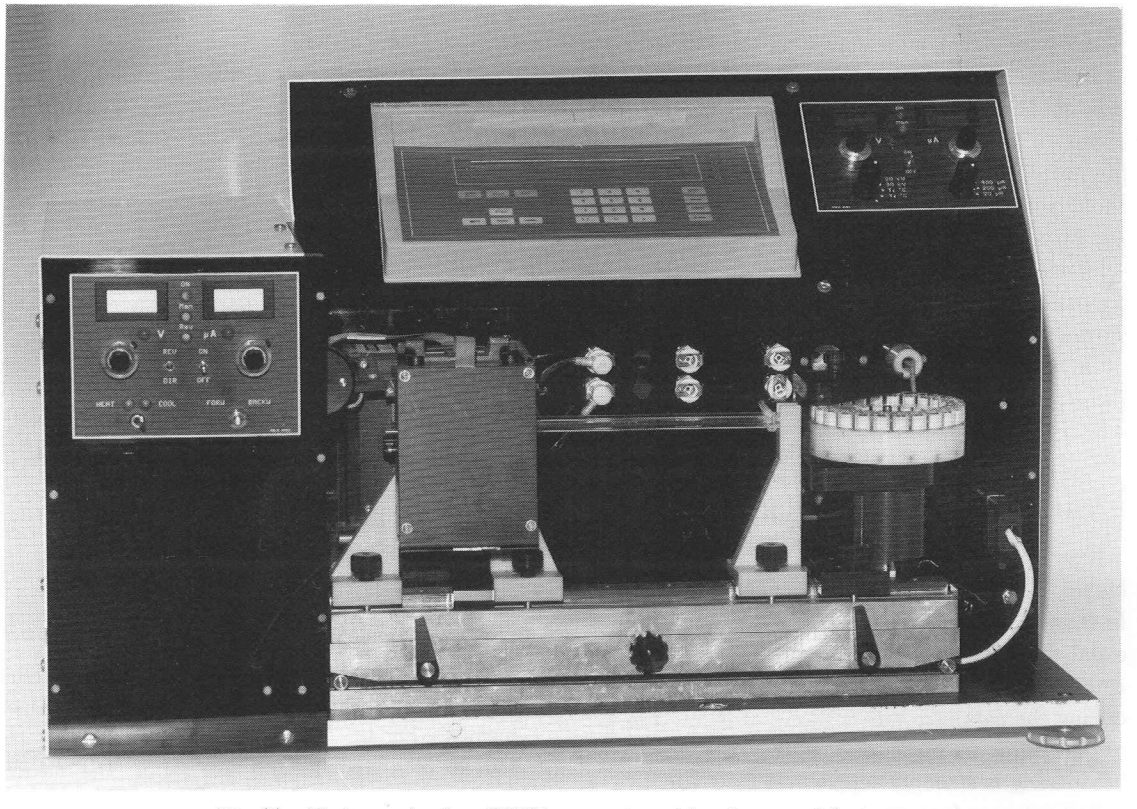


Fig. 14. Photograph of an HPCE apparatus with a few novel features.

HPCE apparatus equipped with a photo-diode array spectrophotometer can display spectra of very small (sample) volumes and thus permits calculation of the above absorption ratio — and thereby the pH.

It is important to know both the pH and the conductivity of the sample to avoid broadening of the starting zone. For adjustment of these parameters, see sections "On-Tube Concentration" and "Off-Tube Concentration".

MICROPREPARATIVE RUNS

In the apparatus designed for such experiments the solutes (proteins) are adsorbed to a moving blotting membrane as they migrate electrophoretically out the capillary. The method permits analysis of the collected, adsorbed proteins (peptides) by mass spectrometry and gas phase sequencing, following washing the membrane with water to remove salts. A similar method is being developed for micro chromatography. Another, older micropreparative method allowing collection of fractions in small test tubes is described in Refs. 23) and 24).

AN HPCE APPARATUS WITH SOME IMPORTANT FEATURES

A photograph of the apparatus is shown in Fig. 14. The capillary is straight since a coiled capillary gives a poorer resolution (see section "The influence of the shape of the capillary on the resolution"). The left electrode vessel is closed to prevent hydrodynamic flow in the capillary (the buffer levels in the electrode vessel need not be the same). The left electrode vessel serves also as an electroendosmotic pump,[25] which permits rinsing of the capillary between the runs. The slit for the UV beam is thermostated to give a stable base line. The Joule heat in the capillary is dissipated efficiently by a rapidly circulating thermostated coolant. The sample is applied electrophoretically in a conventional way or thermally, i.e., by lowering the temperature of the coolant the buffer in the capillary contracts and the sample is sucked in. The length of the starting zone in the latter application mode $=\alpha L^1 \Delta t$, where α=a constant=0.00018 for water, L^1=the length of the capillary and Δt=the temperature decrease.

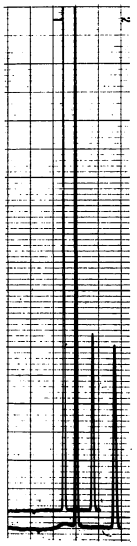


Fig. 15. Application of a narrow starting zone by thermal injection.

Apparatus: see Fig. 14. Sample: benzoic acid and dimethoxybenzoic acid. Dimensions of the capillary: 0.05 (I.D.) ×150 mm. The temperature drop of the coolant from 32°C to 23°C corresponds to a width of only 0.22 mm of the starting zone.

This thermal injection method permits application of very narrow starting zones. An example is given in Fig. 15 (for high resolution, the width of the starting zone should not exceed 1 mm, although a literature search has revealed that much wider starting zones are often employed).

The sample is applied as a minute droplet on the carousel (to the right). The droplet is covered by a cap to prevent evaporation, Just prior to application of the sample the cap is lifted up by a magnet. By effective insulation field strengths up to at least 2,000 V/cm can be applied without current leakage or risk of electric shock.

The apparatus described is still under development.

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